

Smart loan Eligibility Management System

Abstract

The Smart Loan Eligibility & Management System is a database management project developed using MySQL. The system is designed to manage the complete loan application process of a bank or financial institution. It stores customer information, employment details, income, credit score, loan applications, approved loans, EMI schedules, and payment records in a structured database.

Database Creation:

```
CREATE DATABASE smart_loan_system;  
USE smart_loan_system;
```

Table Creation:

```
CREATE TABLE branch (  
    branch_id INT PRIMARY KEY,  
    branch_name VARCHAR(50) NOT NULL,  
    city VARCHAR(50),  
    manager_name VARCHAR(50),  
    contact_no VARCHAR(15)  
);  
-----  
CREATE TABLE customer (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100) NOT NULL,  
    gender VARCHAR(10),  
    dob DATE,  
    phone VARCHAR(15) UNIQUE,  
    email VARCHAR(100) UNIQUE,  
    address VARCHAR(200),  
    branch_id INT,
```

```
    FOREIGN KEY (branch_id) REFERENCES branch(branch_id)
);
```

```
CREATE TABLE employment (
    employment_id INT PRIMARY KEY,
    customer_id INT,
    company_name VARCHAR(100),
    designation VARCHAR(50),
    experience_years INT,
    employment_type VARCHAR(30),
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
);
```

```
CREATE TABLE income (
    income_id INT PRIMARY KEY,
    customer_id INT,
    monthly_income DECIMAL(10,2),
    annual_income DECIMAL(12,2),
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
);
```

```
CREATE TABLE credit_score (
    credit_id INT PRIMARY KEY,
    customer_id INT,
    credit_score INT CHECK (credit_score BETWEEN 300 AND 900),
    last_updated DATE,
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
);
```

```
CREATE TABLE loan_type (
    loan_type_id INT PRIMARY KEY,
    loan_name VARCHAR(50),
    interest_rate DECIMAL(5,2),
    max_amount DECIMAL(12,2)
);
```

```
CREATE TABLE loan_application (  
    application_id INT PRIMARY KEY,  
    customer_id INT,  
    loan_type_id INT,  
    applied_amount DECIMAL(12,2),  
    application_date DATE,  
    application_status VARCHAR(20),  
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id),  
    FOREIGN KEY (loan_type_id) REFERENCES loan_type(loan_type_id)  
);
```

```
CREATE TABLE loan (  
    loan_id INT PRIMARY KEY,  
    application_id INT,  
    approved_amount DECIMAL(12,2),  
    interest_rate DECIMAL(5,2),  
    tenure_months INT,  
    loan_status VARCHAR(20),  
    FOREIGN KEY (application_id) REFERENCES  
loan_application(application_id)  
);
```

```
CREATE TABLE emi (  
    emi_id INT PRIMARY KEY,  
    loan_id INT,  
    emi_amount DECIMAL(10,2),  
    due_date DATE,  
    payment_status VARCHAR(20),  
    FOREIGN KEY (loan_id) REFERENCES loan(loan_id)  
);
```

```
CREATE TABLE payment (  
    payment_id INT PRIMARY KEY,
```

```

emi_id INT,
payment_date DATE,
paid_amount DECIMAL(10,2),
payment_mode VARCHAR(30),
FOREIGN KEY (emi_id) REFERENCES emi(emi_id)
);
-----
-- Insert values
INSERT INTO branch (branch_id, branch_name, city, manager_name, contact_no)
VALUES
(101,'Chennai Main','Chennai','Ramesh','9876543210'),
(102,'Anna Nagar','Chennai','Suresh','9876543211'),
(103,'Tambaram','Chennai','Karthik','9876543212'),
(104,'Velachery','Chennai','Priya','9876543213'),
(105,'OMR','Chennai','Vignesh','9876543214'),
(106,'Coimbatore','Coimbatore','Meena','9876543215'),
(107,'Madurai','Madurai','Arun','9876543216'),
(108,'Trichy','Trichy','Deepa','9876543217'),
(109,'Salem','Salem','Ganesh','9876543218'),
(110,'Erode','Erode','Kavitha','9876543219'),
(111,'Tirunelveli','Tirunelveli','Saravanan','9876543220'),
(112,'Thoothukudi','Thoothukudi','Lakshmi','9876543221'),
(113,'Nagercoil','Nagercoil','Prakash','9876543222'),
(114,'Vellore','Vellore','Divya','9876543223'),
(115,'Cuddalore','Cuddalore','Mohan','9876543224'),
(116,'Kanchipuram','Kanchipuram','Anitha','9876543225'),
(117,'Dindigul','Dindigul','Bala','9876543226'),
(118,'Karur','Karur','Nisha','9876543227'),
(119,'Thanjavur','Thanjavur','Ravi','9876543228'),
(120,'Hosur','Hosur','Sathish','9876543229');
select * from branch;
-----

INSERT INTO customer
(customer_id, customer_name, gender, dob, phone, email, address, branch_id)

```

VALUES

(1001,'Arun
Kumar','Male','1998-05-12','9876500001','arun@gmail.com','Chennai',101),
(1002,'Priya S','Female','1999-08-25','9876500002','priya@gmail.com','Anna
Nagar',102),
(1003,'Karthik
R','Male','1997-11-10','9876500003','karthik@gmail.com','Tambaram',103),
(1004,'Divya
M','Female','2000-01-15','9876500004','divya@gmail.com','Velachery',104),
(1005,'Suresh
B','Male','1996-03-22','9876500005','suresh@gmail.com','OMR',105),
(1006,'Meena
K','Female','1998-07-30','9876500006','meena@gmail.com','Coimbatore',106),
(1007,'Vignesh
P','Male','1995-09-14','9876500007','vignesh@gmail.com','Madurai',107),
(1008,'Anitha
R','Female','1999-06-18','9876500008','anitha@gmail.com','Trichy',108),
(1009,'Ganesh
S','Male','1997-12-09','9876500009','ganesh@gmail.com','Salem',109),
(1010,'Kavitha
T','Female','1998-10-21','9876500010','kavitha@gmail.com','Erode',110),
(1011,'Saravanan
M','Male','1996-02-28','9876500011','saravanan@gmail.com','Tirunelveli',111),
(1012,'Lakshmi
P','Female','1999-04-17','9876500012','lakshmi@gmail.com','Thoothukudi',112),
(1013,'Prakash
V','Male','1995-08-08','9876500013','prakash@gmail.com','Nagercoil',113),
(1014,'Nisha
R','Female','2000-11-27','9876500014','nisha@gmail.com','Vellore',114),
(1015,'Bala
K','Male','1998-01-19','9876500015','bala@gmail.com','Cuddalore',115),
(1016,'Ravi
Kumar','Male','1997-05-06','9876500016','ravi@gmail.com','Kanchipuram',116),
(1017,'Deepa
S','Female','1999-09-12','9876500017','deepa@gmail.com','Dindigul',117),

```

(1018,'Mohan
R','Male','1996-12-24','9876500018','mohan@gmail.com','Karur',118),
(1019,'Harini
V','Female','1998-03-05','9876500019','harini@gmail.com','Thanjavur',119),
(1020,'Sathish
K','Male','1997-07-16','9876500020','sathish@gmail.com','Hosur',120);
select * from customer;
-----
INSERT INTO employment
(employment_id, customer_id, company_name, designation, experience_years,
employment_type)
VALUES
(201,1001,'TCS','Software Engineer',3,'Private'),
(202,1002,'Infosys','System Engineer',2,'Private'),
(203,1003,'Wipro','Java Developer',5,'Private'),
(204,1004,'HCL','Software Developer',2,'Private'),
(205,1005,'Cognizant','Technical Lead',6,'Private'),
(206,1006,'Accenture','Associate Engineer',1,'Private'),
(207,1007,'Zoho','Software Engineer',4,'Private'),
(208,1008,'Capgemini','Senior Developer',5,'Private'),
(209,1009,'Tech Mahindra','Network Engineer',3,'Private'),
(210,1010,'IBM','Database Administrator',7,'Private'),
(211,1011,'State Bank of India','Clerk',8,'Government'),
(212,1012,'Indian Railways','Junior Engineer',4,'Government'),
(213,1013,'L&T','Project Engineer',6,'Private'),
(214,1014,'Amazon','Cloud Engineer',3,'Private'),
(215,1015,'Flipkart','Support Engineer',2,'Private'),
(216,1016,'Tata Motors','Mechanical Engineer',5,'Private'),
(217,1017,'Ashok Leyland','HR Executive',4,'Private'),
(218,1018,'BSNL','Network Administrator',9,'Government'),
(219,1019,'Google','Software Engineer',3,'Private'),
(220,1020,'Microsoft','Full Stack Developer',4,'Private');
select * from employment;
-----

```

```
INSERT INTO income
(income_id, customer_id, monthly_income, annual_income)
VALUES
(301,1001,45000.00,540000.00),
(302,1002,38000.00,456000.00),
(303,1003,65000.00,780000.00),
(304,1004,42000.00,504000.00),
(305,1005,85000.00,1020000.00),
(306,1006,30000.00,360000.00),
(307,1007,70000.00,840000.00),
(308,1008,68000.00,816000.00),
(309,1009,50000.00,600000.00),
(310,1010,90000.00,1080000.00),
(311,1011,40000.00,480000.00),
(312,1012,55000.00,660000.00),
(313,1013,75000.00,900000.00),
(314,1014,80000.00,960000.00),
(315,1015,35000.00,420000.00),
(316,1016,72000.00,864000.00),
(317,1017,48000.00,576000.00),
(318,1018,60000.00,720000.00),
(319,1019,95000.00,1140000.00),
(320,1020,100000.00,1200000.00);
select * from income;
```

```
INSERT INTO credit_score
(credit_id, customer_id, credit_score, last_updated)
VALUES
(401,1001,720,'2026-01-15'),
(402,1002,680,'2026-01-18'),
(403,1003,790,'2026-01-20'),
(404,1004,710,'2026-01-22'),
(405,1005,810,'2026-01-25'),
(406,1006,650,'2026-01-28'),
(407,1007,770,'2026-02-01'),
```

```
(408,1008,730,'2026-02-03'),
(409,1009,690,'2026-02-05'),
(410,1010,840,'2026-02-08'),
(411,1011,700,'2026-02-10'),
(412,1012,760,'2026-02-12'),
(413,1013,820,'2026-02-15'),
(414,1014,780,'2026-02-18'),
(415,1015,660,'2026-02-20'),
(416,1016,740,'2026-02-22'),
(417,1017,710,'2026-02-25'),
(418,1018,750,'2026-02-27'),
(419,1019,850,'2026-03-01'),
(420,1020,800,'2026-03-05');
select * from credit_score;
```

```
INSERT INTO loan_type
(loan_type_id, loan_name, interest_rate, max_amount)
VALUES
(501,'Home Loan',8.50,5000000.00),
(502,'Personal Loan',12.00,1000000.00),
(503,'Vehicle Loan',9.25,1500000.00),
(504,'Education Loan',7.75,2000000.00),
(505,'Gold Loan',10.50,500000.00),
(506,'Business Loan',11.25,3000000.00),
(507,'Agriculture Loan',6.75,1000000.00),
(508,'Home Renovation Loan',9.00,800000.00),
(509,'Medical Loan',10.75,700000.00),
(510,'Travel Loan',13.00,500000.00),
(511,'Wedding Loan',12.50,800000.00),
(512,'Mortgage Loan',8.90,4000000.00),
(513,'Consumer Durable Loan',11.50,300000.00),
(514,'MSME Loan',10.00,2500000.00),
(515,'Startup Loan',9.75,5000000.00),
(516,'Electric Vehicle Loan',8.75,1800000.00),
(517,'Two Wheeler Loan',9.50,200000.00),
```



```
(518,'Commercial Vehicle Loan',10.25,3500000.00),  
(519,'Property Loan',8.80,4500000.00),  
(520,'Loan Against FD',7.25,1000000.00);
```

```
select * from loan_type;
```

```
-----
```

```
INSERT INTO loan_application
```

```
(application_id, customer_id, loan_type_id, applied_amount, application_date,  
application_status)
```

```
VALUES
```

```
(601,1001,501,2500000.00,'2026-01-10','Approved'),  
(602,1002,502,300000.00,'2026-01-12','Rejected'),  
(603,1003,503,800000.00,'2026-01-15','Approved'),  
(604,1004,504,600000.00,'2026-01-18','Pending'),  
(605,1005,505,250000.00,'2026-01-20','Approved'),  
(606,1006,506,1500000.00,'2026-01-22','Rejected'),  
(607,1007,507,500000.00,'2026-01-25','Approved'),  
(608,1008,508,400000.00,'2026-01-28','Pending'),  
(609,1009,509,350000.00,'2026-02-01','Approved'),  
(610,1010,510,200000.00,'2026-02-03','Approved'),  
(611,1011,511,500000.00,'2026-02-06','Rejected'),  
(612,1012,512,3000000.00,'2026-02-08','Approved'),  
(613,1013,513,150000.00,'2026-02-10','Approved'),  
(614,1014,514,1800000.00,'2026-02-12','Pending'),  
(615,1015,515,2200000.00,'2026-02-15','Approved'),  
(616,1016,516,900000.00,'2026-02-18','Approved'),  
(617,1017,517,120000.00,'2026-02-20','Rejected'),  
(618,1018,518,2500000.00,'2026-02-22','Approved'),  
(619,1019,519,3500000.00,'2026-02-25','Approved'),  
(620,1020,520,600000.00,'2026-02-28','Pending');
```

```
select * from loan_application;
```

```
-----
```

```
INSERT INTO loan
```

```
(loan_id, application_id, approved_amount, interest_rate, tenure_months,  
loan_status)
```

VALUES

(701,601,2400000.00,8.50,240,'Approved'),
(702,602,0.00,12.00,0,'Rejected'),
(703,603,750000.00,9.25,60,'Approved'),
(704,604,0.00,7.75,0,'Pending'),
(705,605,240000.00,10.50,36,'Approved'),
(706,606,0.00,11.25,0,'Rejected'),
(707,607,480000.00,6.75,48,'Approved'),
(708,608,0.00,9.00,0,'Pending'),
(709,609,340000.00,10.75,24,'Approved'),
(710,610,190000.00,13.00,24,'Approved'),
(711,611,0.00,12.50,0,'Rejected'),
(712,612,2900000.00,8.90,300,'Approved'),
(713,613,140000.00,11.50,18,'Approved'),
(714,614,0.00,10.00,0,'Pending'),
(715,615,2100000.00,9.75,180,'Approved'),
(716,616,850000.00,8.75,72,'Approved'),
(717,617,0.00,9.50,0,'Rejected'),
(718,618,2450000.00,10.25,120,'Approved'),
(719,619,3400000.00,8.80,240,'Approved'),
(720,620,0.00,7.25,0,'Pending');

select * from loan;

INSERT INTO emi

(emi_id, loan_id, emi_amount, due_date, payment_status)

VALUES

(801,701,23500.00,'2026-03-05','Paid'),
(802,702,0.00,'2026-03-10','Rejected'),
(803,703,15600.00,'2026-03-15','Paid'),
(804,704,0.00,'2026-03-20','Pending'),
(805,705,7200.00,'2026-03-25','Paid'),
(806,706,0.00,'2026-03-28','Rejected'),
(807,707,10500.00,'2026-04-01','Paid'),
(808,708,0.00,'2026-04-05','Pending'),
(809,709,15800.00,'2026-04-10','Paid'),

```
(810,710,9100.00,'2026-04-15','Paid'),  
(811,711,0.00,'2026-04-20','Rejected'),  
(812,712,29800.00,'2026-04-25','Paid'),  
(813,713,8200.00,'2026-05-01','Paid'),  
(814,714,0.00,'2026-05-05','Pending'),  
(815,715,22100.00,'2026-05-10','Paid'),  
(816,716,14600.00,'2026-05-15','Paid'),  
(817,717,0.00,'2026-05-20','Rejected'),  
(818,718,26400.00,'2026-05-25','Pending'),  
(819,719,31200.00,'2026-06-01','Paid'),  
(820,720,0.00,'2026-06-05','Pending');  
select * from emi;
```

```
INSERT INTO payment  
(payment_id, emi_id, payment_date, paid_amount, payment_mode)  
VALUES  
(901,801,'2026-03-05',23500.00,'UPI'),  
(902,802,NULL,0.00,'Not Paid'),  
(903,803,'2026-03-15',15600.00,'Net Banking'),  
(904,804,NULL,0.00,'Not Paid'),  
(905,805,'2026-03-25',7200.00,'Debit Card'),  
(906,806,NULL,0.00,'Not Paid'),  
(907,807,'2026-04-01',10500.00,'UPI'),  
(908,808,NULL,0.00,'Not Paid'),  
(909,809,'2026-04-10',15800.00,'Credit Card'),  
(910,810,'2026-04-15',9100.00,'Cash'),  
(911,811,NULL,0.00,'Not Paid'),  
(912,812,'2026-04-25',29800.00,'Net Banking'),  
(913,813,'2026-05-01',8200.00,'UPI'),  
(914,814,NULL,0.00,'Not Paid'),  
(915,815,'2026-05-10',22100.00,'Debit Card'),  
(916,816,'2026-05-15',14600.00,'UPI'),  
(917,817,NULL,0.00,'Not Paid'),  
(918,818,NULL,0.00,'Pending'),  
(919,819,'2026-06-01',31200.00,'Net Banking'),
```

```
(920,820,NULL,0.00,'Pending');
```

```
select * from payment;
```

```
-----
```

```
-- queries
```

```
SELECT c.customer_name,  
       i.monthly_income  
FROM customer c  
INNER JOIN income i  
ON c.customer_id = i.customer_id  
WHERE i.monthly_income >  
(  
    SELECT AVG(monthly_income)  
    FROM income  
);
```

```
-----
```

```
SELECT c.customer_name,  
       l.approved_amount  
FROM customer c  
JOIN loan_application la  
ON c.customer_id = la.customer_id  
JOIN loan l  
ON la.application_id = l.application_id  
WHERE c.customer_id IN  
(  
    SELECT customer_id  
    FROM credit_score  
    WHERE credit_score > 750  
);
```

```
-----
```

```
DELIMITER $$
```

```
CREATE FUNCTION calculate_emi(  
    amount DECIMAL(12,2),  
    months INT
```

```
)  
RETURNS DECIMAL(10,2)  
DETERMINISTIC  
BEGIN  
    IF months = 0 THEN  
        RETURN 0;  
    END IF;  
  
    RETURN amount / months;  
END $$
```

```
DELIMITER  
drop function calculate_emi;  
SELECT loan_id,  
       approved_amount,  
       tenure_months,  
       calculate_emi(approved_amount, tenure_months) AS EMI  
FROM loan;  
-----  
DELIMITER $$
```

```
CREATE FUNCTION loan_category(  
    amount DECIMAL(12,2)  
)  
RETURNS VARCHAR(20)  
DETERMINISTIC  
BEGIN  
    DECLARE category VARCHAR(20);  
  
    IF amount >= 2000000 THEN  
        SET category = 'High Loan';  
    ELSEIF amount >= 500000 THEN  
        SET category = 'Medium Loan';  
    ELSE  
        SET category = 'Low Loan';
```

END IF;

RETURN category;
END \$\$

DELIMITER ;
SELECT loan_id,
approved_amount,
loan_category(approved_amount) AS Loan_Category
FROM loan;

DELIMITER \$\$

CREATE PROCEDURE GetAllCustomers()
BEGIN
SELECT * FROM customer;
END \$\$

DELIMITER ;
CALL GetAllCustomers();

DELIMITER \$\$

CREATE PROCEDURE GetLoanDetails()
BEGIN
SELECT c.customer_name,
lt.loan_name,
l.approved_amount,
l.loan_status
FROM customer c
JOIN loan_application la
ON c.customer_id = la.customer_id
JOIN loan_type lt
ON la.loan_type_id = lt.loan_type_id
JOIN loan l

```
        ON la.application_id = l.application_id;  
END $$
```

```
DELIMITER ;  
CALL GetLoanDetails();
```

```
-----
```

```
DELIMITER $$
```

```
CREATE TRIGGER check_credit_score  
BEFORE INSERT  
ON credit_score  
FOR EACH ROW  
BEGIN  
    IF NEW.credit_score < 300 OR NEW.credit_score > 900 THEN  
        SIGNAL SQLSTATE '45000'  
        SET MESSAGE_TEXT='Invalid Credit Score';  
    END IF;  
END $$
```

```
DELIMITER ;  
INSERT INTO credit_score  
VALUES(421,1001,950,'2026-07-10');  
DELIMITER $$
```

```
CREATE TRIGGER check_loan_amount  
BEFORE UPDATE  
ON loan  
FOR EACH ROW  
BEGIN  
    IF NEW.approved_amount < 0 THEN  
        SIGNAL SQLSTATE '45000'  
        SET MESSAGE_TEXT='Loan Amount Cannot Be Negative';  
    END IF;  
END $$
```

DELIMITER ;

```
UPDATE loan
SET approved_amount=-10000
WHERE loan_id=701;
DELIMITER $$
```

```
CREATE TRIGGER update_emi_status
AFTER INSERT
ON payment
FOR EACH ROW
BEGIN
    UPDATE emi
    SET payment_status='Paid'
    WHERE emi_id=NEW.emi_id;
END $$
```

```
DELIMITER ;
INSERT INTO payment
VALUES(921,818,'2026-07-10',26400,'UPI');
SELECT *
FROM emi
WHERE emi_id=818;
-----
```

```
SELECT c.customer_name,
       l.approved_amount,
       l.loan_status
FROM customer c
JOIN loan_application la
ON c.customer_id = la.customer_id
JOIN loan l
ON la.application_id = l.application_id
WHERE c.customer_id IN
(
```



```
SELECT customer_id
FROM customer
WHERE customer_name LIKE 'A%'
);
-----
SELECT b.branch_name,
       COUNT(c.customer_id) AS total_customers
FROM branch b
JOIN customer c
ON b.branch_id = c.branch_id
JOIN income i
ON c.customer_id = i.customer_id
WHERE i.monthly_income >
(
  SELECT AVG(monthly_income)
  FROM income
)
GROUP BY b.branch_name
HAVING COUNT(c.customer_id) >= 1;
-----
```

```
SELECT c.customer_name,
       l.approved_amount
FROM customer c
JOIN loan_application la
ON c.customer_id = la.customer_id
JOIN loan l
ON la.application_id = l.application_id
WHERE l.approved_amount >
(
  SELECT AVG(approved_amount)
  FROM loan
)
ORDER BY l.approved_amount DESC
LIMIT 5;
```

